Reinforcement learning

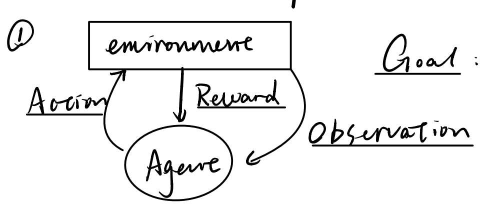
"Learning through experience to make good decision under uncertainty"

"Reinforcement learning"

$\Updownarrow$

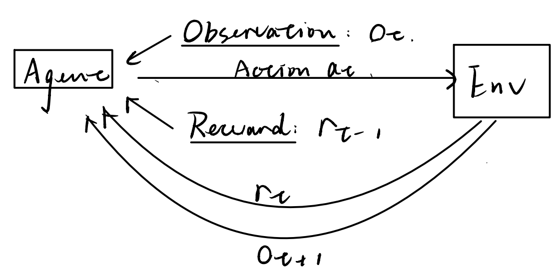
"Supervised learning": learning through expert demonstration

Basic RL Concept

① 

Goal: Maximize total expected reward

At each step t:



② History and State

History: a sequence: $O_1, a_1, r_1, O_2, a_2 \dots, r_{t-1}, O_t, a_t, r_t$

"All observable variables up to time t"

State: "the information used to determine what happens next"

"$S_t = f(h_t)$" ⇒ depends on the history

Agent State: S_t^a "information that agent have"

$S_t^a = f(h_t)$. f can be any function

③ Reward Hypothesis

"All goals can be described by the maximization of expected cumulative reward"

④ Markov State

$p(S_{t+1} \mid S_t) = p(S_{t+1} \mid S_t \dots S_1)$ => Also. $p(h_{t+1} \mid h_t) = p(h_{t+1} \mid h_t \dots h_1)$

⇒ State S_{t+1} ONLY depends on S_t

⇒ future is independent of past given the present

In practice. S_t often equals to O_t

⑤ Fully Observable Environment

→ "$S_t^e = S_t^a$" Agent State = Environment State

→ ⑥ MDP "Markov decision process" = Markov State + Fully Observable Env

$\Updownarrow$

POMDP "Partially Observable Markov decision process"

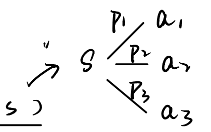
= Markov State + Partially Observable ENV. ($S_t^e \neq S_t^a$)

RL Agent

1. Policy: agent's behavior function
2. Value Function: how good is each state and/or action
3. Model: Agent's representation of the environment

Policy: Deterministic policy: $a = \pi(s)$

Stochastic policy $\pi(a \mid s) = p(a_t = a \mid S_t = s)$



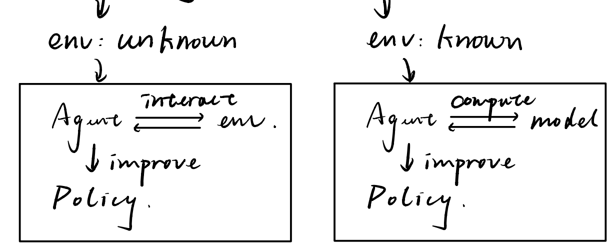
Value Function: $V_\pi(s) = \underbrace{\mathbb{E}_\pi}_{\text{expectation}}[r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} \dots \mid S_t = s]$, where γ is a discount factor

Model: predicts what environment will do next

$P(s' \mid s, a) = \underbrace{P(S_{t+1} = s' \mid S_t = s, a_t = a)}_{\text{Statistical}}$

$r(s, a) = \mathbb{E}[r_t \mid S_t = s, a_t = a]$

Learning vs. Planning



Exploration: Find more info about env.

Exploitation: Exploit known info to maximize reward.

Characteristic of RL

① No Supervision.

② Feedback is delayed.

③ Time matters.

④ Action affects the subsequent data

⑤ Active learning (collect own data)

	AI Planning	Supervised Learning	Unsupervised Learning	RL	Imitation Learning
Policy Optimization	✓			✓	✓
Learning from Experience		✓	✓	✓	✓
Generalization	✓	✓	✓	✓	✓
Delayed Consequences	✓			✓	✓
Exploration				✓	